25900805 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCECT102 - DATA ANALYTICS

(2024 ADMISION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **35** Questions

Each Question carries **1** mark

1. What is an effective way to mitigate risks associated with multimedia data? [U] / [CO1]

   a). Reducing the amount of multimedia data used
   b). Regularly updating data formats and ensuring security measures are in place
   c). Ignoring data quality assessments
   d). None of the above

2. Name one outcome of effectively using big data in organizations. [U] / [CO1]

   a). Decreased customer satisfaction
   b). Enhanced operational efficiency
   c). Higher data redundancy
   d). Increased manual labor

3. Identify a scenario where secondary data is used. [U] / [CO1]

a). Conducting a new experiment to test a hypothesis	b). Reviewing historical documents for trends	c). Collecting firsthand data through surveys	d). Interviewing participants for a study

4. Which of the following best describes quantitative data? [K] / [CO1]

a). Data that describes attributes or qualities	b). Data that is subjective and based on opinions	c). Data that can be measured and expressed numerically	d). Data that cannot be categorized into numbers

5. Select the process used to remove inaccuracies or inconsistencies from raw data. [U] / [CO1]

a). Data storage	b). Data cleaning	c). Data visualization	d). Data exporting

6. Which sector has significantly benefited from data analytics in recent years? [K] / [CO1]

a). Only healthcare	b). Only retail	c). All sectors, including healthcare, finance, and marketing	d). Only manufacturing

7. What is one way data analytics can enhance customer satisfaction? [U] / [CO1]

a). By reducing communication with customers	b). By personalizing customer experiences	c). By limiting product choices	d). By ignoring customer complaints

8. In what way is numerical data typically used in predictive analytics? [U] / [CO1]

a). By determining qualitative trends in the market	b). By providing insights into customer satisfaction	c). By identifying relationships between variables and making forecasts	d). By categorizing individuals into groups

9. Which of the following is an example of qualitative data? [K] / [CO1]

a). Height of a person	b). Distance between two cities.	c). Product satisfaction level (e.g., "satisfied", "dissatisfied").	d). Monthly income.

10. Identify a scenario where secondary data is used. [U] / [CO1]

a). Conducting a new experiment to test a hypothesis	b). Reviewing historical documents for trends	c). Collecting firsthand data through surveys	d). Interviewing participants for a study

11. Recognize the relationship between discrete data and counting. [U] / [CO1]

a). Discrete data is not related to counting	b). Discrete data comes from counting individual units	c). Discrete data involves continuous measurement	d). Discrete data must be measured using a ruler

12. Identify how navigation data can be visualized in descriptive analytics. [U] / [CO1]

a). By forecasting new market trends	b). By displaying real-time GPS updates for vehicles	c). By generating future traffic flow simulations	d). By creating heat maps of historical traffic congestion and movement patterns

13. It is true that effective interpretation of data in predictive analytics requires: [U] / [CO1]

a). Ignoring historical data patterns	b). A focus solely on qualitative analysis	c). An understanding of business context and objectives	d). Only statistical knowledge

14. Recognize a key characteristic of secondary data. [U] / [CO1]

a). It is collected directly by the researcher	b). It is collected by someone else for a different purpose	c). It is always qualitative	d). It is less reliable than primary data

15. When exporting data to a PDF, which option allows you to specify what part of the workbook to export? [U] / [CO1]

a). Print Area	b). Export Selection	c). Save As	d). Export All

16. Recognize a type of variable that is discrete. [U] / [CO1]

a). The number of books in a library	b). The speed of a car	c). The time spent running	d). The volume of water in a bottle

17. If you want to save a copy of the current workbook in a different location, which command should you use? [U] / [CO1]

a). Open b). Save c). Save As d). Export

18. Name the type of data that primary data refers to. [K] / [CO2]

a). Data collected from secondary sources b). Data obtained from published reports c). Data collected firsthand by the researcher d). Data sourced from the internet

19. How do diagnostic and prescriptive analytics differ fundamentally? [A] / [CO1]

a). They use the same techniques b). Diagnostic analytics focuses on "what happened" while prescriptive focuses on "what should be done" c). Prescriptive analytics is only concerned with historical data d). There is no difference; they are interchangeable terms

20. What action would you take to modify a scatter plot to better display a relationship between two variables? [A] / [CO1]

a). Add a trendline to the scatter plot b). Change the scatter plot to a pie chart c). Increase the number of data points in the scatter plot d). Add bars to the plot

21. Which of the following tools or techniques is commonly used for de-duplication in databases? [K] / [CO1]

a). Data Encryption b). Hashing algorithms c). Random Sampling d). Feature Engineering

22. When creating a histogram, which action will provide more detailed insight into the distribution of your data? [U] / [CO1]

a). Increase the bin width to group more data points together b). Decrease the bin width to create more intervals for better precision c). Use fewer bins to simplify the chart d). Convert the histogram into a bar chart

23. If you have a pie chart showing department expenditures, and you want to display the percentage of total for each category, what feature would you use? [A] / [CO1]

a). Add data labels and choose the option to show percentages	b). Change the pie chart to a bar chart	c). Adjust the colour scheme of the chart	d). Insert a line connecting each category to its percentage

24. Which of the following steps would you take to visualize monthly temperature data over the past year using a Line Chart in Excel? [K] / [CO1]

a). Create a pie chart and select the temperature data.	b). Insert a line chart and choose the monthly temperature data as the source.	c). Use conditional formatting on the temperature data.	d). Sort the data by temperature.

25. Which action would you take to make a bar graph easier to interpret if it has too many categories? [U] / [CO1]

a). Increase the chart size	b). Group similar categories or limit the number of bars displayed	c). Convert the chart into a pie chart	d). Use different colours for each bar

26. You are working with a dataset where feature A has a range between 10 and 100, and feature B has a range between 0.01 and 0.1. Which step would you apply to make the features comparable? [A] / [CO1]

a). Drop feature A to balance the data	b). Standardize both features so they have a mean of 0 and standard deviation of 1	c). Replace all values in feature B with values from feature A	d). Perform one-hot encoding on feature A

27. What is the primary goal of data preprocessing? [K] / [CO1]

a). To increase the size of the dataset	b). To clean and prepare raw data for analysis	c). To generate new variables	d). To visualize the data

28. When importing data from a text file, which delimiter is commonly used? [K] / [CO1]

a). Semicolon	b). Comma	c). Space	d). Both A and B

29. You want to display a column chart showing the number of units sold by different sales representatives. How can you ensure that the chart dynamically updates as you add more sales data? [A] / [CO1]

a). Use a static chart range | b). Use a named range or Excel table to automatically adjust the chart data | c). Manually update the chart after adding data | d). Increase the chart size

30. Which of the following is an advantage of using surveys for data collection? [K] / [CO1]

a). They are expensive to conduct | b). They provide quick access to large amounts of data | c). They guarantee high response rates | d). They only allow for in-person administration

31. You want to add a line of best fit to your scatter chart to make predictions based on your data. Which of the following should you do? [A] / [CO1]

a). Adjust the axis scale | b). Insert a trendline that fits the data | c). Increase the size of the data markers | d). Convert the scatter chart into a line chart

32. Which formatting option allows you to shade cells based on their values relative to others? [K] / [CO1]

a). Color Scales | b). Highlight Cells Rules | c). Icon Sets | d). Custom Rules

33. What is the result of =LOWER("Pampa river")? [A] / [CO2]

a). Pampa River | b). Pampa river | c). PAMPA RIVER | d). pampa river

34. If cell A1 contains "apple,banana", then the result of formula=SUBSTITUTE(A1,"orange","apple") [A] / [CO2]

a). apple,orange | b). orange,banana | c). apple,banana | d). None of the above

35. Which of the following is a key benefit of using visualizations in data analysis? [U] / [CO2]

a). They replace the need for statistical analysis | b). They can reveal trends and patterns quickly | c). They limit the audience's understanding of data | d). They are only useful for large datasets

36. Which function returns the sum from a set of values? [U] / [CO2]

a). MIN | b). MAX | c). AVERAGE | d). SUM

37. Which function can be used to get the number of empty cells in a range? [U] / [CO1]

a). COUNTA	b). COUNT	c). COUNTIF	d). COUNTBLANK

38. You are conducting an experiment to test the effectiveness of a new learning method. What should you apply to ensure unbiased results? [A] / [CO1]

a). Use convenience sampling for participants	b). Randomly assign participants to experimental and control groups	c). Allow participants to choose their group	d). Disregard the control group

39. Which of the following is a key feature of focus groups? [K] / [CO1]

a). One-on-one discussion	b). Large-scale data analysis	c). Group interaction and discussion	d). Pre-defined coding schemes

40. What is the main reason for converting data types during the cleaning process? [U] / [CO1]

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[35X1=35]

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[35X1=35]

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